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Electric work machine

Abstract

An electric work machine is driven appropriately. An electric work machine includes a motor including a stator, a rotor rotatable relative to the stator, and a rotor shaft fixed to the rotor, an output unit drivable by the rotor shaft, a motor case accommodating the stator and the rotor, a magnet located inside the motor case and rotatable by the rotor, and a magnetic sensor located inside the motor case to detect the magnet.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application No. 2021-207415, filed on Dec. 21, 2021, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an electric work machine.

2. Description of the Background

(3) In the field of electric work machines, a known power tool includes a motor and a sensor board, as described in Japanese Unexamined Patent Application Publication No. 2019-017129.

BRIEF SUMMARY

- (4) A motor includes a stator and a rotor rotatable relative to the stator. The rotation of the rotor is detected by magnetic sensors supported on a sensor board. The motor is controlled based on detection signals obtained by the magnetic sensors. The electric work machine may not be driven appropriately when the detection accuracy of the magnetic sensors is reduced.
 - (5) One or more aspects of the present disclosure are directed to an electric work machine that is driven appropriately.
 - (6) A first aspect of the present disclosure provides an electric work machine, including: a motor including a stator, a rotor rotatable relative to the stator, and a rotor shaft fixed to the rotor; an output unit drivable by the rotor shaft; a motor case accommodating the stator and the rotor; a magnet located inside the motor case and rotatable by the rotor; and a magnetic sensor located inside the motor case to detect the magnet.
 - (7) The electric work machine according to the above aspect of the present disclosure is driven appropriately.
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Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a left perspective view of an electric work machine according to a first embodiment.
- (2) FIG. 2 is a left perspective view of a motor assembly in the first embodiment.
- (3) FIG. 3 is a side view of the motor assembly in the first embodiment.
- (4) FIG. 4 is a longitudinal sectional view of the motor assembly in the first embodiment.
- (5) FIG. 5 is a cross-sectional view of the motor assembly in the first embodiment.
- (6) FIG. 6 is a partially enlarged longitudinal sectional view of the motor assembly in the first embodiment.
- (7) FIG. 7 is a partially enlarged longitudinal sectional view of a motor assembly in a second embodiment.
- (8) FIG. 8 is an exploded perspective view of a sensor board and a heat insulating plate in the second embodiment.
- (9) FIG. 9 is a partially enlarged longitudinal sectional view of a motor assembly in a third embodiment.
- (10) FIG. 10 is a longitudinal sectional view of a motor assembly in a fourth embodiment.
- (11) FIG. 11 is a partially enlarged longitudinal sectional view of the motor assembly in the fourth embodiment.
- (12) FIG. 12 is a partially enlarged longitudinal sectional view of a motor assembly in a fifth embodiment.
- (13) FIG. 13 is a left perspective view of a lid in the fifth embodiment.
- (14) FIG. 14 is a right perspective view of the lid in the fifth embodiment.

DETAILED DESCRIPTION

- (15) Although one or more embodiments of the present disclosure will now be described with reference to the drawings, the present disclosure is not limited to the present embodiments. The components in the embodiments described below may be combined as appropriate. One or more components may be eliminated.
- (16) In the embodiments, the positional relationships between the components will be described using the directional terms such as right and left (or lateral), front and rear (or frontward and rearward), and up and down (or vertical). The terms indicate relative positions or directions with respect to the center of an electric work machine 1.
- (17) The electric work machine 1 includes a motor 12. A direction parallel to a rotation axis AX of the motor 12 is referred to as an axial direction for convenience. A direction radial from the rotation

axis AX of the motor **12** is referred to as a radial direction or radially for convenience. A direction about the rotation axis AX of the motor **12** is referred to as a circumferential direction, circumferentially, or a rotation direction for convenience.

(18) A position nearer the center of the motor **12** in the axial direction, or an axial direction toward the center of the motor **12**, is referred to as axially inward for convenience. A position farther from the center of the motor **12** in the axial direction, or an axial direction away from the center of the motor **12**, is referred to as axially outward for convenience. A position nearer the rotation axis AX of the motor **12** in the radial direction, or a radial direction toward the rotation axis AX, is referred to as radially inward for convenience. A position farther from the rotation axis AX of the motor **12** in the radial direction, or a radial direction away from the rotation axis AX, is referred to as radially outside or radially outward for convenience. A position in one circumferential direction, or one circumferential direction, is referred to as a first circumferential direction for convenience. A position in the other circumferential direction, or the other circumferential direction, is referred to as a second circumferential direction for convenience.

First Embodiment

(19) A first embodiment will now be described.

(20) Electric Work Machine

(21) FIG. **1** is a left perspective view of the electric work machine **1** according to the present embodiment. The electric work machine **1** according to the present embodiment is a chain saw as an example of outdoor power equipment.

(22) The electric work machine **1** includes a housing **2**, a front grip **3**, a hand guard **4**, a battery cover **5**, a controller **6**, a trigger lock lever **7**, a trigger switch **8**, a guide bar **9**, a saw chain **10**, and a motor assembly **11A**.

(23) The housing **2** is formed from a synthetic resin. The housing **2** includes a motor compartment **14**, a battery holder **15**, and a rear grip **16**.

(24) The motor compartment **14** accommodates the motor assembly **11A**. The motor assembly **11A** includes the motor **12**.

(25) The battery holder **15** holds a battery pack (not shown). The battery holder **15** includes a battery mount to which the battery pack is attached. The battery holder **15** holds the battery pack with the battery mount in between. The battery holder **15** is connected to the rear end of the motor compartment **14**. The battery cover **5** covers the battery pack held in the battery holder **15**.

(26) The battery pack is detachable from the battery holder **15**. The battery pack includes a secondary battery. The battery pack in the present embodiment includes a rechargeable lithium-ion battery. The battery pack functions as a power supply for the electric work machine **1**. The battery pack is held in the battery holder **15** to power the electric work machine **1**.

(27) The rear grip **16** is grippable by a hand of a user of the electric work machine **1**. The rear grip **16** is connected to the rear end of the battery holder **15**. A portion of the rear grip **16** is connected to an upper portion of the rear end of the battery holder **15**. A portion of the rear grip **16** is connected to a lower portion of the rear end of the battery holder **15**.

(28) The motor compartment **14** has an inlet port **2A** in its left portion. The motor compartment **14** has an outlet port **2B** in its front portion. Air outside the housing **2** flows into the housing **2** through the inlet port **2A**. Air inside the housing **2** flows out of the housing **2** through the outlet port **2B**.

(29) The front grip **3** is grippable by a hand of the user of the electric work machine **1**. The front grip **3** is formed from a synthetic resin. The front grip **3** is a pipe. The front grip **3** is connected to the housing **2**. The front grip **3** has its left end connected to the left side surface of the motor compartment **14**. The front grip **3** has its right end connected to the right side surface of the battery holder **15**.

(30) The hand guard **4** protects the hand of the user holding the front grip **3**. The hand guard **4** is located in front of the front grip **3**. The hand guard **4** is connected to an upper portion of the motor compartment **14**.

- (31) The controller **6** outputs control signals for controlling the electric work machine **1**. The controller **6** controls a drive current supplied from the battery pack to the motor **12**. The controller **6** is accommodated in the motor compartment **14**.
- (32) The trigger lock lever **7** is operable by the user of the electric work machine **1** to allow the trigger switch **8** to be operable. The trigger lock lever **7** is located on the rear grip **16**.
- (33) The trigger switch **8** is operable by the user of the electric work machine **1** to drive the motor **12**. The trigger switch **8** is located on the rear grip **16**. The user of the electric work machine **1** holding the rear grip **16** with the hand operates the trigger lock lever **7** to operate the trigger switch **8** with a finger. A drive current is supplied to the motor **12** to drive the motor **12**.
- (34) The guide bar **9** guides the saw chain **10**. The guide bar **9** is a plate elongated in the front-rear direction. The guide bar **9** extends frontward from the housing **2**.
- (35) The saw chain **10** functions as an output unit of the electric work machine **1** driven by the motor **12**. The saw chain **10** includes multiple cutters that are connected to one another. The saw chain **10** is located along the peripheral edge of the guide bar **9**. The motor **12** and the saw chain **10** are connected with a power transmission (not shown) including a sprocket. In response to the trigger switch **8** being operated to rotate the motor **12**, the saw chain **10** moves along the peripheral edge of the guide bar **9**.
- (36) Motor Assembly
- (37) FIG. **2** is a left perspective view of the motor assembly **11A** in the present embodiment. FIG. **3** is a side view of the motor assembly **11A** in the present embodiment. FIG. **4** is a longitudinal sectional view of the motor assembly **11A** in the present embodiment. FIG. **5** is a cross-sectional view of the motor assembly **11A** in the present embodiment. FIG. **6** is a partially enlarged longitudinal sectional view of the motor assembly **11A** in the present embodiment.
- (38) The motor assembly **11A** includes the motor **12**, a motor case **13**, bearings **18**, a heat-transfer resin portion **19**, a cooling fan **20**, a rotary member **64**, a sensor board **22**, a seal **23**, a press assembly **24**, seals **25**, and a seal **26**.
- (39) The motor **12** is a power source for the electric work machine **1**. The motor **12** is an electric motor drivable with a drive current supplied from the battery pack.
- (40) The motor **12** is a brushless inner-rotor motor. The motor **12** includes a stator **27**, a rotor **28**, and a rotor shaft **29**. The rotor **28** rotates relative to the stator **27**. The rotor shaft **29** is fixed to the rotor **28**. The stator **27** surrounds at least a part of the rotor **28**. The rotor **28** rotates about the rotation axis AX. The saw chain **10** is driven by the rotor shaft **29**.
- (41) The rotor **28** in the present embodiment has the rotation axis AX extending laterally. The axial direction parallel to the rotation axis AX of the rotor **28** corresponds to the lateral direction.
- (42) The stator **27** includes a stator core **30**, an insulator **31**, coils **32**, and a busbar unit **33**. The stator core **30** includes multiple steel plates stacked on one another. The steel plates are metal plates formed from iron as a main component. The stator core **30** includes an annular yoke **34** and teeth **35**. The teeth **35** protrude radially inward from the inner surface of the yoke **34**. The yoke **34** surrounds the rotation axis AX. Multiple (**12** in the present embodiment) teeth **35** are located at circumferentially equal intervals.
- (43) The insulator **31** covers at least a part of the surface of the stator core **30**. The insulator **31** covers at least the outer peripheral surfaces of the teeth **35**. The insulator **31** is formed from a synthetic resin. The insulator **31** is fixed to the stator core **30**. The insulator **31** is integral with the stator core **30**. The insulator **31** is fixed to the stator core **30** by insert molding.
- (44) The coils **32** are wound around the teeth **35** with the insulator **31** in between. The coils **32** and the stator core **30** are insulated from each other by the insulator **31**. Multiple (**12** in the present embodiment) coils **32** are located circumferentially.
- (45) The busbar unit **33** is fastened to the insulator **31** with screws **53**. A drive current from the battery pack is supplied to the busbar unit **33** through the controller **6**.
- (46) The busbar unit **33** includes external terminals **42**, short-circuiting members **44**, and an

insulating member **45**.

(47) The external terminals **42** are connected to the short-circuiting members **44** and to connection terminals **47**. The external terminals **42** are connected to the coils **32** with the short-circuiting members **44**. Each short-circuiting member **44** connects (short-circuits) a pair of coils **32** radially opposed to each other. The external terminals **42** are connected to the connection terminals **47**, power lines **46**, and the battery pack with the controller **6**. Each external terminal **42** and the corresponding connection terminal **47** are fastened with a screw **52**. The connection terminals **47** are connected to the power lines **46**. The power lines **46** are lead wires connected to the coils **32** with the connection terminals **47**, the external terminals **42**, and the short-circuiting members **44**. A drive current from the battery pack is supplied to the coils **32** through the controller **6**, the power lines **46**, the connection terminals **47**, the external terminals **42**, and the short-circuiting members **44**.

(48) The insulating member **45** is formed from a synthetic resin. The insulating member **45** surrounds the rotation axis AX. The short-circuiting members **44** are located inside the insulating member **45**. The insulating member **45** supports the external terminals **42** and the short-circuiting members **44**.

(49) The controller **6** controls the drive current supplied from the battery pack to the busbar unit **33**. A drive current from the battery pack is supplied to the external terminals **42** in the busbar unit **33** through the controller **6**, the power lines **46**, and the connection terminals **47**. The drive current supplied from the battery pack to the external terminals **42** flows through the short-circuiting members **44**, and is then supplied to the coils **32**.

(50) The rotor **28** includes a rotor core **54** and rotor magnets **55**.

(51) The rotor core **54** includes multiple steel plates stacked on one another. The steel plates are metal plates formed from iron as a main component. The rotor core **54** surrounds the rotation axis AX. The stator core **30** surrounds the rotor core **54**.

(52) The rotor magnets **55** are permanent magnets. The rotor magnets **55** are held by the rotor core **54**. The rotor magnets **55** are located inside the rotor core **54**. Multiple (eight in the present embodiment) rotor magnets **55** are located circumferentially.

(53) The rotor core **54** has multiple magnet slots **56** located circumferentially at intervals. The magnet slots **56** extend through the right and left end faces of the rotor core **54**. The rotor magnets **55** are received in the respective magnet slots **56**.

(54) The rotor core **54** in the present embodiment includes multiple (four in the present embodiment) hollow portions **57** located circumferentially at intervals. The hollow portions **57** extend through the right and the left end faces of the rotor core **54**. The hollow portions **57** are located radially inward from the magnet slots **56**. The hollow portions **57** reduce the weight of the rotor core **54**.

(55) The rotor shaft **29** extends in the axial direction. The rotor shaft **29** has the central axis aligned with the rotation axis AX. The rotor shaft **29** is located inward from the rotor core **54**. The rotor core **54** is fixed to the rotor shaft **29**. In the present embodiment, a cylindrical member **58** surrounds the rotor shaft **29**. The cylindrical member **58** is formed from an electrically insulating material. The rotor shaft **29** is fixed to the rotor core **54** with the cylindrical member **58** in between. The rotor shaft **29** includes a left portion protruding leftward from the left end face of the rotor core **54**. The rotor shaft **29** includes a right portion protruding rightward from the right end face of the rotor core **54**.

(56) A drive current is supplied to the coils **32** from the battery pack through the controller **6** to generate a rotating magnetic field in the stator **27**. This causes the rotor **28** and the rotor shaft **29** to rotate about the rotation axis AX.

(57) The motor case **13** accommodates at least a part of the motor **12**. The motor case **13** in the present embodiment accommodates at least the stator **27** and the rotor **28**. The motor case **13** has an internal space accommodating the stator **27** and the rotor **28**. The stator **27** surrounds at least a part

of the rotor **28** in the internal space of the motor case **13**. The internal space is closed. The internal space in the present embodiment is sealed substantially hermetically. The rotor shaft **29** is partially located in the internal space.

(58) The motor case **13** is formed from a metal. The motor case **13** in the present embodiment is formed from aluminum. The motor case **13** may be formed from an aluminum die-casting alloy (ADC12).

(59) The motor case **13** includes a body **60**, heat-radiating fins **61**, and a lid **62**.

(60) The body **60** accommodates the stator **27** and the rotor **28**. The stator **27** and the rotor **28** are located inside the body **60**.

(61) The body **60** includes a cylindrical portion **60A**, an extension **60B**, and a wall **60C**. The cylindrical portion **60A** surrounds the rotation axis AX. The extension **60B** protrudes frontward from the cylindrical portion **60A**. The wall **60C** is connected to the right end of the cylindrical portion **60A**. The body **60** and the extension **60B** have an opening **63** in their left ends. The stator **27** is placed inside the body **60** through the opening **63**.

(62) The heat-radiating fins **61** are located on the outer surface of the cylindrical portion **60A**. The heat-radiating fins **61** protrude radially outward from the outer surface of the cylindrical portion **60A**. The heat-radiating fins **61** extend in the axial direction on the outer surface of the cylindrical portion **60A**. Multiple heat-radiating fins **61** are located circumferentially at intervals.

(63) The lid **62** is fastened to the body **60** to cover the opening **63**. This defines an internal space between the body **60** and the lid **62**. At least a part of the motor **12** including the stator **27** is placed inside the body **60**, and then the opening **63** is covered with the lid **62**. The motor **12** is thus at least partially located in the internal space.

(64) The body **60** includes multiple (four in the present embodiment) screw bosses **69** located circumferentially at intervals about the opening **63**. Each screw boss **69** has a threaded hole. The lid **62** includes multiple (four in the present embodiment) screw bosses **70** located at intervals on its peripheral edge. Each screw boss **70** has an opening. A screw **71** has its distal end fastened into the threaded hole in the corresponding screw boss **69**, with its middle portion received in the opening of the screw boss **70**. This structure fastens the body **60** and the lid **62**.

(65) The lid **62** includes a disk **62A** and an extension **62B**. The disk **62A** is connected to the cylindrical portion **60A**. The extension **62B** is connected to the extension **60B**. The extension **62B** protrudes frontward from the disk **62A**.

(66) The bearings **18** support the rotor shaft **29**. The rotor shaft **29** is rotatably supported by the bearings **18**. The bearings **18** are supported on the motor case **13**.

(67) The motor case **13** has shaft holes **72** that at least partially receive the rotor shaft **29**. The shaft holes **72** connect the internal space and the external space of the motor case **13**. The shaft holes **72** include a shaft hole **72L** and a shaft hole **72R**. The shaft hole **72L** receives the left portion of the rotor shaft **29**. The shaft hole **72R** receives the right portion of the rotor shaft **29**. The shaft hole **72L** is located in the disk **62A** in the lid **62**. The shaft hole **72R** is located in the wall **60C** of the body **60**.

(68) The motor case **13** includes peripheral walls **101** defining the shaft holes **72**. The peripheral walls **101** are substantially cylindrical. The peripheral walls **101** include a peripheral wall **101L** and a peripheral wall **101R**. The peripheral wall **101L** defines the shaft hole **72L**. The peripheral wall **101R** defines the shaft hole **72R**. The peripheral wall **101L** is located on the disk **62A** in the lid **62**. The peripheral wall **101L** protrudes leftward from the left surface of the disk **62A**. The peripheral wall **101R** is located on the wall **60C** of the body **60**. The peripheral wall **101R** protrudes rightward from the right surface of the wall **60C**.

(69) The bearings **18** include a left bearing **18L** and a right bearing **18R**. The left bearing **18L** supports the left portion of the rotor shaft **29**. The right bearing **18R** supports the right portion of the rotor shaft **29**. The left bearing **18L** is received in the shaft hole **72L**. The right bearing **18R** is received in the shaft hole **72R**.

(70) The right and left ends of the rotor shaft **29** are located outside the motor case **13**. The right end of the rotor shaft **29** is connected to the saw chain **10** with the power transmission (not shown) including the sprocket. The saw chain **10** is driven by the rotor shaft **29**. The rotor shaft **29** rotates to drive the saw chain **10**.

(71) The heat-transfer resin portion **19** is accommodated in the motor case **13**. The heat-transfer resin portion **19** is in contact with the coils **32** and the motor case **13**. The heat-transfer resin portion **19** covers the coils **32**. The heat-transfer resin portion **19** in the present embodiment is in contact with the stator core **30** and the insulator **31**.

(72) The heat-transfer resin portion **19** is formed from a synthetic resin. The heat-transfer resin portion **19** is highly thermally conductive and electrically insulating. For example, when the insulator **31** is formed from a nylon resin with a thermal conductivity of $0.2 \text{ W}/(\text{m}\cdot\text{K})$, the thermal conductivity of the synthetic resin used for the heat-transfer resin portion **19** is higher than $0.2 \text{ W}/(\text{m}\cdot\text{K})$.

(73) An insulating synthetic resin with a thermal conductivity higher than $0.2 \text{ W}/(\text{m}\cdot\text{K})$ is, for example, an unsaturated polyester resin. The heat-transfer resin portion **19** may be formed from a nylon resin containing insulating, thermally conductive fillers.

(74) After the stator **27** is placed inside the body **60** through the opening **63**, a heated and melted synthetic resin is fed inside the body **60** through the opening **63**. The synthetic resin fed inside the body **60** solidifies to form the heat-transfer resin portion **19**.

(75) The cooling fan **20** is located outside the motor case **13**. The cooling fan **20** faces at least a part of the outer surface of the motor case **13**. The cooling fan **20** is fixed to the rotor shaft **29**. The cooling fan **20** is rotated by the rotor shaft **29**.

(76) The cooling fan **20** in the present embodiment is located leftward from the motor case **13**. The cooling fan **20** is fixed to the left end of the rotor shaft **29** located outside the motor case **13**. The cooling fan **20** faces at least a part of the outer surface of the motor case **13**. The cooling fan **20** in the present embodiment faces at least a part of the lid **62**.

(77) The cooling fan **20** in the present embodiment is a centrifugal fan. The cooling fan **20** has an inflow port **20A** and an outflow port **20F**. The inflow port **20A** is located at the left end of the cooling fan **20**. The outflow port **20F** is located on the periphery of the cooling fan **20**. The cooling fan **20** generates an airflow to cool the motor case **13**. As the rotor shaft **29** rotates, the cooling fan **20** rotates together with the rotor shaft **29**. Air around the cooling fan **20** is then drawn through the inflow port **20A**. The air drawn through the inflow port **20A** is blown out through the outflow port **20F**. At least a part of the air blown out through the outflow port **20F** hits the outer surface of the motor case **13**. The motor case **13** is thus cooled.

(78) The cooling fan **20** in the present embodiment is fixed to the rotor shaft **29** with a fan bush **73**. The fan bush **73** is an intermediate member connecting the rotor shaft **29** and the cooling fan **20**. The fan bush **73** is located outside the motor case **13** to connect the left end of the rotor shaft **29** and the cooling fan **20**. As the cooling fan **20** rotates, the fan bush **73** rotates together with the cooling fan **20**.

(79) The rotary member **64** supports sensor magnets **21**. The rotary member **64** is located inside the motor case **13**. The rotary member **64** is fixed to the rotor shaft **29** inside the motor case **13**.

(80) In the present embodiment, a sleeve **113** surrounds the rotor shaft **29**. The sleeve **113** is fixed to the rotor shaft **29**. The sleeve **113** is formed from a metal such as brass. The sleeve **113** functions as a rotational balancer that corrects any rotational imbalance of the rotor **28**.

(81) The rotary member **64** surrounds the sleeve **113**. The rotary member **64** is fixed to the sleeve **113**. The rotary member **64** is fixed to the rotor shaft **29** with the sleeve **113** in between. As the rotor shaft **29** rotates, the rotary member **64** rotates together with the rotor shaft **29**.

(82) The sensor magnets **21** are fixed to the rotary member **64**. The sensor magnets **21** are permanent magnets. Multiple (eight in the present embodiment) sensor magnets **21** are located circumferentially at intervals. The sensor magnets **21** and the rotor magnets **55** are at the same

positions in the circumferential direction.

(83) The sensor magnets **21** are located inside the rotary member **64**. The rotary member **64** in the present embodiment has multiple magnet slots **65** located circumferentially at intervals. The magnet slots **65** extend through the right and left end faces of the rotary member **64**. The sensor magnets **21** are received in the respective magnet slots **65**.

(84) The sensor magnets **21** are rotated by the rotor **28**. As the rotary member **64** is rotated by the rotor **28** and the rotor shaft **29**, the sensor magnets **21** rotate.

(85) The sensor board **22** supports magnetic sensors **74**. The sensor board **22** is located inside the motor case **13**. The sensor board **22** is annular. The sensor board **22** surrounds the rotor shaft **29**. The sensor board **22** includes a printed wiring board (PWB). The sensor board **22** is formed from, for example, an epoxy resin. The sensor board **22** may be formed from a metal, such as aluminum, or from a heat-radiating resin with a higher thermal conductivity than an epoxy resin.

(86) In the axial direction, the sensor board **22** is located between at least a part of the motor case **13** and the sensor magnets **21**. The sensor board **22** is fixed to the motor case **13**. The sensor board **22** is fastened to the lid **62** with screws **78**.

(87) The magnetic sensors **74** detect the sensor magnets **21**. Each magnetic sensor **74** is, for example, a Hall device. The magnetic sensors **74** detect the sensor magnets **21** to detect rotation of the rotor **28**. Five signal lines **75** are connected to the sensor board **22**. The magnetic sensors **74** transmit detection signals to the controller **6** through the signal lines **75**. The five signal lines **75** in the present embodiment are bundled together with a tube **115**.

(88) In the axial direction, the rotary member **64** is located between the sensor board **22** and the rotor **28**. The magnetic sensors **74** are supported by the sensor board **22** to face the sensor magnets **21**.

(89) In the present embodiment, a resin film **76** covers the magnetic sensors **74** and at least a part of the surface of the sensor board **22**. The resin film **76** functions as a heat insulating film.

(90) The motor case **13** includes a wiring path **79** connecting the internal space and the external space of the motor case **13**. The wiring path **79** is defined in the extension **60B** of the body **60**. The wiring path **79** has an insertion end opening **80** facing the outer space of the motor case **13**. The insertion end opening **80** is located in the front end of the wiring path **79**.

(91) The power lines **46** are received in the wiring path **79**. The power lines **46** are connected to the coils **32** with the connection terminals **47** and the busbar unit **33**.

(92) The seal **23** seals between each power line **46** and the motor case **13**. The seal **23** seals between the tube **115** and the motor case **13**. The seal **23** is substantially cylindrical. The seal **23** is formed from rubber. The seal **23** is received in the wiring path **79**. The seal **23** is placed into the wiring path **79** through the insertion end opening **80**.

(93) The seal **23** has four holes **81** receiving the three power lines **46** and a single tube **115**. The four holes **81** are parallel to one another. Each hole **81** receives the corresponding power line **46**. One hole **81** receives the tube **115**. The outer surface of each power line **46** and the inner surface of the corresponding hole **81** are in close contact with each other. The outer surface of the tube **115** and the inner surface of the hole **81** are in close contact with each other.

(94) The press assembly **24** applies pressure to the seal **23** received in the wiring path **79**. The press assembly **24** compresses the seal **23** to cause the inner surface of each hole **81** in the seal **23** to be in close contact with the outer surface of the corresponding power line **46**. The press assembly **24** compresses the seal **23** to cause the outer surface of the tube **115** and the inner surface of the hole **81** to be in close contact with each other. The press assembly **24** presses the seal **23** against the motor case **13** to cause the seal **23** and at least a part of the motor case **13** to be in close contact with each other.

(95) The press assembly **24** includes a press member **91** and a cover **92**. The press member **91** includes a ring **93** and multiple tabs **94**. The ring **93** surrounds the seal **23**. The multiple tabs **94** are connected to the ring **93**.

(96) The ring **93** surrounds the seal **23**. The tabs **94** are connected to a front portion of the ring **93**. The multiple tabs **94** surround the seal **23**. The tabs **94** are elastically deformable to move toward each other. In other words, the press member **91** is elastically deformable to have a smaller diameter at the tabs **94**.

(97) With the cover **92** in contact with the press member **91**, the cover **92** is fixed to the motor case **13**. The cover **92** includes a peripheral wall **95** and fasteners **96**. The peripheral wall **95** surrounds the tabs **94**. The fasteners **96** are fixed to the motor case **13**.

(98) With the inner surface of the peripheral wall **95** and the tabs **94** in contact with each other, the cover **92** is fastened to the motor case **13** with screws **99**. The cover **92** then moves backward toward the motor case **13**. The peripheral wall **95** causes the multiple tabs **94** to elastically deform to move toward each other. The multiple tabs **94** thus compress the seal **23**. This causes the outer surface of the power line **46** to come in close contact with the inner surface of the hole **81**, and the outer surface of the tube **115** to come in close contact with the inner surface of the hole **81**.

(99) When the cover **92** moves backward, the press member **91** moves backward together with the cover **92**. This causes the seal **23** to move backward together with the press member **91** and be pressed against at least a portion of the motor case **13**. The seal **23** and the motor case **13** are thus in close contact with each other.

(100) The outer surface of each power line **46** and the inner surface of the corresponding hole **81** being in close contact with each other seal between each power line **46** and the seal **23**. The outer surface of the tube **115** and the inner surface of the hole **81** being in close contact with each other seal between the tube **115** and the seal **23**. The seal **23** and the motor case **13** being in close contact with each other seal between the seal **23** and the motor case **13**. Thus, the seal **23** seals between each power line **46** and the motor case **13** and between the tube **115** and the motor case **13**.

(101) The seals **25** seal between the rotor shaft **29** and the motor case **13**. The seals **25** include oil seals. The seals **25** are received in the shaft holes **72**. The seals **25** are press-fitted between the rotor shaft **29** and the peripheral walls **101**. The seals **25** are supported by the peripheral walls **101**. The seals **25** include a seal **25L** and a seal **25R**. The seal **25L** seals between the left portion of the rotor shaft **29** and the motor case **13**. The seal **25R** seals between the right portion of the rotor shaft **29** and the motor case **13**. The seal **25L** is received in the shaft hole **72L**. The seal **25R** is received in the shaft hole **72R**.

(102) A stopper **108L** is located at the left of the seal **25L**. A stopper **108R** is located at the right of the seal **25R**. The stopper **108L** prevents the seal **25L** from slipping out of the shaft hole **72L**. The stopper **108R** prevents the seal **25R** from slipping out of the shaft hole **72R**. The stopper **108L** and the stopper **108R** are, for example, circlips.

(103) The seal **26** seals between the body **60** and the lid **62**. The seal **26** includes an O-ring. The body **60** has a recess **109** surrounding the opening **63**. The seal **26** is received in the recess **109**.

(104) As described above, the electric work machine **1** according to the present embodiment includes the motor **12** and the saw chain **10** that is an output unit. The motor **12** includes the stator **27**, the rotor **28** rotatable relative to the stator **27**, and the rotor shaft **29** fixed to the rotor **28**. The saw chain **10** is driven by the rotor shaft **29**. The electric work machine **1** includes the motor case **13** accommodating the stator **27** and the rotor **28**. The electric work machine **1** includes the sensor magnets **21** located inside the motor case **13** and rotatable by the rotor **28**. The electric work machine **1** includes the sensor board **22** located inside the motor case **13** and supporting the magnetic sensors **74** that detect the sensor magnets **21**.

(105) In the above structure, the sensor magnets **21** are rotated by the rotor **28**, and the magnetic sensors **74** then detect the sensor magnets **21** to detect rotation of the rotor **28**. The sensor magnets **21** and the sensor board **22** inside the motor case **13** are protected by the motor case **13**. The detection accuracy of the magnetic sensors **74** is thus less likely to be reduced.

(106) For example, for the sensor magnets **21** outside the motor case **13**, magnetic materials such as iron powder around the motor case **13** may adhere to the sensor magnets **21**. For the sensor board

22 outside the motor case 13, the magnetic sensors 74 may deteriorate. The sensor magnets 21 with such magnetic materials or the deteriorating magnetic sensors 74 may reduce the detection accuracy of the magnetic sensors 74. The motor case 13 protects the sensor magnets 21 and the sensor board 22. The detection accuracy of the magnetic sensors 74 is thus less likely to be reduced. The controller 6 controls the motor 12 based on detection signals obtained by the magnetic sensors 74 that have the detection accuracy less likely to be reduced. The electric work machine 1 is thus driven appropriately.

(107) In the present embodiment, the sensor board 22 is located between at least a part of the motor case 13 and the sensor magnets 21 in the axial direction parallel to the rotation axis AX of the rotor 28.

(108) This structure can avoid a size increase of the motor assembly 11A including the motor case 13, the sensor magnets 21, and the sensor board 22.

(109) In the present embodiment, the sensor board 22 is fixed to the motor case 13.

(110) This structure reduces the likelihood that the relative positions between the motor case 13 and the sensor board 22 change.

(111) The motor case 13 in the present embodiment includes the body 60 having the opening 63 and accommodating the stator 27 and the rotor 28, and the lid 62 fastened to the body 60 and covering the opening 63. The sensor board 22 is fastened to the lid 62.

(112) With the lid 62 to which the sensor board 22 is fastened is fastened to the body 60, the sensor board 22 is accommodated inside the motor case 13.

(113) The electric work machine 1 according to the present embodiment includes the rotary member 64 fixed to the rotor shaft 29 inside the motor case 13. The sensor magnets 21 detected by the magnetic sensors 74 are fixed to the rotary member 64.

(114) In the above structure, the sensor magnets 21 fixed to the rotary member 64 are detected by the magnetic sensors 74. The magnetic sensors 74 are thus less susceptible to heat from the rotor 28. As the motor 12 is driven, the rotor 28 may reach high temperature and generate heat. When the rotor magnets 55 are detected by the magnetic sensors 74, the sensor board 22 is to be placed near the rotor 28. Heat from the rotor 28 may thus disable the magnetic sensors 74 from operating appropriately. The rotary member 64 is less likely to reach high temperature. Although the sensor board 22 is located adjacent to the rotary member 64, the magnetic sensors 74 detect the sensor magnets 21 and can thus operate appropriately. For the structure in which the rotary member 64 and the rotor 28 are both inside the motor case 13, the sensor magnets 21 and the rotor magnets 55 can be magnetized simultaneously in the process of producing the sensor magnets 21 and the rotor magnets 55. More specifically, the rotary member 64 and the rotor 28 fixed on the rotor shaft 29 can be together placed into a magnetizer. The sensor magnets 21 and the rotor magnets 55 are less likely to be misaligned in the rotational direction. The controller 6 can control the motor 12 appropriately based on detection signals obtained by the magnetic sensors 74 detecting the sensor magnets 21.

(115) In the present embodiment, the rotary member 64 is located between the sensor board 22 and the rotor 28 in the axial direction parallel to the rotation axis AX of the rotor 28.

(116) This increases the distance between the sensor board 22 and the rotor 28. The magnetic sensors 74 are thus less susceptible to heat from the rotor 28. The rotary member 64 between the sensor board 22 and the rotor 28 blocks radiant heat from the rotor 28 and reduces heat transfer to the sensor board 22. The magnetic sensors 74 thus operate appropriately.

(117) The sensor board 22 in the present embodiment may be formed from a metal or a heat-radiating resin.

(118) The sensor board 22 formed from the heat-radiating material can reduce a temperature increase in the sensor board 22 when heat from the rotor 28 is transferred to the sensor board 22. The sensor board 22 is connected to the motor case 13 to cause heat in the sensor board 22 to be transferred to the motor case 13 and radiate from the motor case 13. The magnetic sensors 74 thus

operate appropriately.

(119) The electric work machine **1** according to the present embodiment includes the cooling fan **20** located outside the motor case **13** and rotatable by the rotor shaft **29**.

(120) The rotor shaft **29** rotates the cooling fan **20** to cool the motor case **13**. The sensor board **22** is less likely to increase temperature. The magnetic sensors **74** thus operate appropriately. The motor case **13** is cooled to reduce a temperature increase in the motor **12**. The motor **12** thus operates appropriately. This also reduces the likelihood that electronic devices accommodated in the motor case **13** malfunction under heat or electronic components accommodated in the motor case **13** deteriorate under heat. The electric work machine **1** is thus driven appropriately.

(121) The cooling fan **20** in the present embodiment is fixed to the end of the rotor shaft **29** located outside the motor case **13**.

(122) This structure allows the cooling fan **20** to rotate together with the rotor shaft **29** outside the motor case **13**.

(123) The cooling fan **20** in the present embodiment faces at least a part of the outer surface of the motor case **13**.

(124) In the above structure, the cooling fan **20** can efficiently direct air onto the outer surface of the motor case **13**.

Second Embodiment

(125) A second embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.

(126) FIG. **7** is a partially enlarged longitudinal sectional view of a motor assembly **11B** in the present embodiment. As shown in FIG. **7**, the motor assembly **11B** includes the sensor board **22** located inside the motor case **13**. The sensor board **22** supports the magnetic sensors **74**. As in the first embodiment, the sensor board **22** may be formed from an epoxy resin, a metal, or a heat-radiating resin with a higher thermal conductivity than an epoxy resin.

(127) The motor assembly **11B** in the present embodiment does not include the rotary member **64** and the sensor magnets **21** unlike in the first embodiment. The magnetic sensors **74** detect the rotor magnets **55** rotated by the rotor **28**. The rotor **28** including the rotor magnets **55** is located inside the motor case **13**.

(128) The motor assembly **11B** in the present embodiment includes a heat insulating plate **760** covering the magnetic sensors **74** and at least a part of the surface of the sensor board **22**. The heat insulating plate **760** is formed from a synthetic resin.

(129) FIG. **8** is an exploded perspective view of the sensor board **22** and the heat insulating plate **760** in the present embodiment. The heat insulating plate **760** includes a first heat insulating plate **760A** and a second heat insulating plate **760B**. The first heat insulating plate **760A** covers the left surface of the sensor board **22**. The second heat insulating plate **760B** covers the right surface of the sensor board **22**. The magnetic sensors **74** are located on the right surface of the sensor board **22**. The sensor board **22** includes, in addition to the magnetic sensors **74**, multiple electronic components **77** on its right surface. The electronic components **77** mounted on the sensor board **22** are, for example, a capacitor, a resistor, a diode, and a thermistor. The second heat insulating plate **760B** covers the magnetic sensors **74** and the electronic components **77**.

(130) As described above, the motor assembly **11B** in the present embodiment includes the heat insulating plate **760** covering the magnetic sensors **74** and at least a part of the surface of the sensor board **22**.

(131) In the above structure, heat from the rotor **28** is blocked by the heat insulating plate **760** and thus is restricted from being transferred to the magnetic sensors **74** and to the sensor board **22**. The magnetic sensors **74** and the sensor board **22** are also protected by the heat insulating plate **760**. The magnetic sensors **74** thus operate appropriately.

Third Embodiment

(132) A third embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.

(133) FIG. **9** is a partially enlarged longitudinal sectional view of a motor assembly **11C** in the present embodiment. As shown in FIG. **9**, the motor assembly **11C** includes the sensor board **22** located inside the motor case **13**. The sensor board **22** supports the magnetic sensors **74**. As in the first embodiment, the sensor board **22** may be formed from an epoxy resin, a metal, or a heat-radiating resin with a higher thermal conductivity than an epoxy resin.

(134) The motor assembly **11C** includes a heat insulator **770** between the sensor board **22** and the rotor **28** in the axial direction. The heat insulator **770** is, for example, a plate formed from a synthetic resin. The heat insulator **770** is fixed to the left end face of the rotor **28** facing the sensor board **22**.

(135) As described above, the motor assembly **11C** in the present embodiment includes the heat insulator **770** between the sensor board **22** and the rotor **28** in the axial direction parallel to the rotation axis AX of the rotor **28**.

(136) In the above structure, heat from the rotor **28** is blocked by the heat insulator **770** and thus is restricted from being transferred to the sensor board **22**. The magnetic sensors **74** thus operate appropriately.

(137) The heat insulator **770** in the present embodiment is fixed to the left end face of the rotor **28** facing the sensor board **22**.

(138) Thus, the heat insulator **770** rotates together with the rotor **28**. The motor assembly **11C** can also avoid a size increase.

Fourth Embodiment

(139) A fourth embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.

(140) FIG. **10** is a longitudinal sectional view of a motor assembly **11D** in the present embodiment. FIG. **11** is a partially enlarged longitudinal sectional view of the motor assembly **11D** in the present embodiment. As shown in FIGS. **10** and **11**, the motor assembly **11D** includes the sensor board **22** located inside the motor case **13**. The sensor board **22** supports the magnetic sensors **74**. As in the first embodiment, the sensor board **22** may be formed from an epoxy resin, a metal, or a heat-radiating resin with a higher thermal conductivity than an epoxy resin.

(141) The motor assembly **11D** includes stirring fans **220** located inside the motor case **13** and rotatable by the rotor shaft **29**. The stirring fans **220** are fixed to the rotor shaft **29** inside the motor case **13**. The stirring fans **220** include a stirring fan **220L** and a stirring fan **220R**. The stirring fan **220L** is fixed to the left portion of the rotor shaft **29**. The stirring fan **220R** is fixed to the right portion of the rotor shaft **29**. As the rotor shaft **29** rotates, the stirring fans **220** rotate together with the rotor shaft **29** in the motor case **13**. This stirs the air inside the motor case **13**.

(142) As described above, the motor assembly **11D** includes the stirring fans **220** inside the motor case **13** and rotatable by the rotor shaft **29**.

(143) With the air inside the motor case **13** being stirred by the stirring fans **220**, heat from the air is more easily transferred to the motor case **13**. The heat transferred to the motor case **13** radiates from the motor case **13**. This reduces a temperature increase in the motor case **13**. The magnetic sensors **74** thus operate appropriately.

(144) The stirring fans **220** in the present embodiment are fixed to the rotor shaft **29** inside the motor case **13**.

(145) The stirring fans **220** can thus rotate together with the rotor shaft **29** inside the motor case **13**.

Fifth Embodiment

(146) A fifth embodiment will now be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described

briefly or will not be described.

(147) FIG. 12 is a partially enlarged longitudinal sectional view of a motor assembly 11E in the present embodiment. FIG. 13 is a left perspective view of a lid 620 in the present embodiment. FIG. 14 is a right perspective view of the lid 620 in the present embodiment.

(148) As shown in FIG. 12, the motor assembly 11E includes the sensor board 22 located inside the motor case 13. The sensor board 22 supports the magnetic sensors 74. As in the first embodiment, the sensor board 22 may be formed from an epoxy resin, a metal, or a heat-radiating resin with a higher thermal conductivity than an epoxy resin.

(149) As shown in FIG. 14, the motor assembly 11E includes internal heat-radiating fins 612 located on the inner surface of the motor case 13. The motor case 13 has its inner surface facing the internal space of the motor case 13. The internal heat-radiating fins 612 in the present embodiment are located on the inner surface of the lid 620. The internal heat-radiating fins 612 protrude rightward from the inner surface (right surface) of the lid 620. The internal heat-radiating fins 612 extend in the radial direction. Multiple internal heat-radiating fins 612 are located circumferentially at intervals.

(150) As shown in FIG. 13, the lid 620 includes external heat-radiating fins 611. The external heat-radiating fins 611 are located on the outer surface of the lid 620. The external heat-radiating fins 611 protrude leftward from the outer surface (left surface) of the lid 620. The external heat-radiating fins 611 extend in the radial direction. Multiple external heat-radiating fins 611 are located circumferentially at intervals.

(151) As described above, the motor assembly 11E in the present embodiment includes the internal heat-radiating fins 612 on the inner surface of the motor case 13.

(152) The motor case 13 is efficiently cooled by the internal heat-radiating fins 612.

(153) The internal heat-radiating fins 612 in the present embodiment are located on the inner surface of the lid 620.

(154) The internal heat-radiating fins 612 are located on the lid 620 facing the cooling fan 20. The motor case 13 is thus cooled efficiently.

(155) The motor assembly 11E in the present embodiment includes the external heat-radiating fins 611 on the outer surface of the lid 620.

(156) The external heat-radiating fins 611 are located on the lid 620 facing the cooling fan 20. The motor case 13 is thus cooled efficiently.

OTHER EMBODIMENTS

(157) In each of the above embodiments, the magnetic sensors 74 are located on the sensor board 22. The magnetic sensors 74 may not be located on the sensor board 22. The magnetic sensors 74 may be located directly on a resin member or connected directly to a lead wire (signal line).

(158) In the above embodiments, the motor 12 is a brushless inner-rotor motor. The motor 12 may be a brushless outer-rotor motor. In the brushless outer-rotor motor, the teeth protrude radially outward from the annular yoke.

(159) The electric work machine 1 according to the above embodiments is a chain saw, which is an example of outdoor power equipment. The outdoor power equipment may not be a chain saw. Examples of the outdoor power equipment include a hedge trimmer, a lawn mower, a mowing machine, and a blower. The electric work machine 1 may be a power tool. Examples of the power tool include a driver drill, a vibration driver drill, an angle drill, an impact driver, a grinder, a hammer, a hammer drill, a circular saw, and a reciprocating saw.

(160) In the above embodiments, the electric work machine is powered by the battery pack attached to the battery mount. In some embodiments, the electric work machine may use utility power (alternating current power supply).

REFERENCE SIGNS LIST

(161) 1 electric work machine 2 housing 2A inlet port 2B outlet port 3 front grip 4 hand guard 5 battery cover 6 controller 7 trigger lock lever 8 trigger switch 9 guide bar 10 saw chain (output

unit) 11A motor assembly 11B motor assembly 11C motor assembly 11D motor assembly 11E motor assembly 12 motor 13 motor case 14 motor compartment 15 battery holder 16 rear grip 18 bearing 18L left bearing 18R right bearing 19 heat-transfer resin portion 20 cooling fan 20A inflow port 20F outflow port 21 sensor magnet 22 sensor board 23 seal 24 press assembly 25 seal 25L seal 25R seal 26 seal 27 stator 28 rotor 29 rotor shaft 30 stator core 31 insulator 32 coil 33 busbar unit 34 yoke 35 tooth 42 external terminal 44 short-circuiting member 45 insulating member 46 power line 47 connection terminal 52 screw 53 screw 54 rotor core 55 rotor magnet 56 magnet slot 57 hollow portion 58 cylindrical member 60 body 60A cylindrical portion 60B extension 60C wall 61 heat-radiating fin 62 lid 62A disk 62B extension 63 opening 64 rotary member 65 magnet slot 69 screw boss 70 screw boss 71 screw 72 shaft hole 72L shaft hole 72R shaft hole 73 fan bush 74 magnetic sensor 75 signal line 76 resin film 77 electronic component 78 screw 79 wiring path 80 insertion end opening 81 hole 91 press member 92 cover 93 ring 94 tab 95 peripheral wall 96 fastener 99 screw 101 peripheral wall 101L peripheral wall 101R peripheral wall 108L stopper 108R stopper 109 recess 113 sleeve 115 tube 220 stirring fan 220L stirring fan 220R stirring fan 611 external heat-radiating fin 612 internal heat-radiating fin 620 lid 760 heat insulating plate 760A first heat insulating plate 760B second heat insulating plate 770 heat insulator

Claims

1. An electric work machine, comprising: a housing formed from a synthetic resin; a motor including a stator, a rotor rotatable relative to the stator, and a rotor shaft fixed to the rotor; an output unit drivable by the rotor shaft; a motor case having a hermetically sealed internal space accommodating the stator and the rotor, the motor case accommodated in the housing, the motor case formed from a metal; a magnet located inside the motor case and rotatable by the rotor; and a magnetic sensor located inside the motor case to detect the magnet.
2. The electric work machine according to claim 1, further comprising: a sensor board supporting the magnetic sensor configured to be cooled by a cooling fan rotated by the rotor shaft.
3. The electric work machine according to claim 2, wherein the sensor board is located between at least a part of the motor case and the magnet in an axial direction parallel to a rotation axis of the rotor.
4. The electric work machine according to claim 2, wherein the sensor board is fixed to the motor case.
5. The electric work machine according to claim 4, wherein the motor case includes a body having an opening and accommodating the stator and the rotor, and a lid fastened to the body and covering the opening, and the sensor board is fastened to the lid.
6. The electric work machine according to claim 2, wherein the sensor board comprises a metal or a heat-radiating resin.
7. The electric work machine according to claim 1, further comprising: a rotary member fixed to the rotor shaft inside the motor case, wherein the magnet detectable by the magnetic sensor is a sensor magnet fixed to the rotary member.
8. The electric work machine according to claim 7, wherein the rotary member is located between a sensor board and the rotor in an axial direction parallel to a rotation axis of the rotor.
9. The electric work machine according to claim 1, further comprising: a heat insulating plate covering the magnetic sensor.
10. The electric work machine according to claim 1, further comprising: a heat insulator between the magnetic sensor and the rotor in an axial direction parallel to a rotation axis of the rotor.
11. The electric work machine according to claim 10, wherein the heat insulator is fixed to an end face of the rotor facing the magnetic sensor.
12. The electric work machine according to claim 1, further comprising: a stirring fan located inside the motor case and rotatable by the rotor shaft.

13. The electric work machine according to claim 12, wherein the stirring fan is fixed to the rotor shaft inside the motor case.
 14. The electric work machine according to claim 1, further comprising: a cooling fan located outside the motor case and rotatable by the rotor shaft.
 15. The electric work machine according to claim 14, wherein the cooling fan is fixed to an end of the rotor shaft located outside the motor case.
 16. The electric work machine according to claim 14, wherein the cooling fan faces at least a part of an outer surface of the motor case.
 17. The electric work machine according to claim 1, further comprising: an internal heat-radiating fin on an inner surface of the motor case.
 18. The electric work machine according to claim 3, wherein the sensor board is fixed to the motor case.
 19. The electric work machine according to claim 3, wherein the sensor board comprises a metal or a heat-radiating resin.
 20. The electric work machine according to claim 4, wherein the sensor board comprises a metal or a heat-radiating resin.
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